

DIVYAPRAKASH

+91-78749 80691 divyaprakash.poddar@gmail.com in divyaprakash-iitd divyaprakash-iitd dpcfd.com

EDUCATION

Year	Degree / Board	Institute	GPA / Marks(%)
2025 (Expected)	Ph.D.	Indian Institute of Technology Delhi	9
2016	M.Tech.	Indian Institute of Technology Gandhinagar	8.11
2012	B.E.	Dr. Ambedkar Institute of Technology, Bangalore	76

TECHNICAL SKILLS

- Programming:** Python, MATLAB, Fortran, C/C++ (proficient); Git, Bash (experienced); Web development (basic)
- Modeling & Simulation:** OpenFOAM, ANSYS Fluent, Star-CCM+, ParaView, gmsk (proficient/experienced); SU2, FDS (experienced)
- CAD:** SolidWorks, FreeCAD, Blender (experienced)
- AI/ML:** PyTorch, TensorFlow (experienced); machine learning for scientific computing (applications to CFD)

SELECTED PROJECTS

- Computational & Experimental Modeling of Fire Suppression Systems (2016–2017)** : Developed detonation and deflagration simulations using Uintah and conducted fire suppression experiments modeled in FDS, highlighting combustion dynamics and heat transfer processes. 📄 Slides & Animations
- Towards High-Fidelity Modeling of Superdroplet Condensation using DNS Data and Machine Learning (2023 – 2025)**: Developed a machine learning-based parameterization for droplet growth in LES using high-fidelity DNS data. 📄 Paper 🔄 Code

📁 Full Project Portfolio

INTERNSHIPS

- Google Summer of Code 2025, Remote (May - Sept, 2025)** : Stitching SU2 – Using data-driven, physics-informed machine learning to model fluid properties in CFD. 📄 Report
- Institute of High Performance Computing, Singapore (May – July, 2015)** : Implementation of immersed boundary method in OpenFOAM

WORK EXPERIENCE

- Jorhat Engineering College, Assam, India, Assistant Professor, (Sept, 2018 – Sept, 2020)**
- Institute of Technology Gopeshwar, Uttarakhand, India, Assistant Professor, (Jan, 2018 – Aug, 2018)**
- Indian Institute of Technology Gandhinagar, Gujarat, India, Junior Research Fellow, (Sept, 2016 – Aug, 2017)**
- Institute for Plasma Research Gandhinagar, Gujarat, India, Graduate Apprentice, (Aug, 2013 – Nov, 2013)**

IIT DELHI PHD THESIS

Title: Sensing and Transport of Particles Using Arrays of Artificial Cilia

Supervisor: Amitabh Bhattacharya

Description: Cilia, hair-like organelles for fluid transport, locomotion, and sensing, are modeled via Kirchhoff rods and particles via FEM, in a 2D immersed boundary framework of cilia–fluid–deformable particle interactions and extended to a 3D OpenFOAM coupled setup. A novel, non-optical method using an LSTM neural network infers particle size and shape from cilia deflections, and simulations show how ciliary peristalsis drives particle transport.

SELECTED PUBLICATIONS

- A Review of Computational Modeling of Fluid-Immersed Flexible Filaments**, *Journal of the Indian Institute of Science*, 2024. DOI: 10.1007/s41745-024-00423-x
- Machine Learning Based Sensing of Particle Shape and Size Using Passive Artificial Cilia**, *Journal of Fluids and Structures* (under review), 2025.

🔗 Google Scholar (ID: H4Byf9EAAAAJ)

CONFERENCES

- preCICE Workshop 2024, University of Stuttgart, Germany (Sept 2024)**: Poster Presentation – Awarded Travel Grant
- ECCOMAS Congress 2024, Lisbon, Portugal (June 2024)**: Oral Presentation – Art & Science Competition Winner
- Workshop on Interfacial Engineering at Multiple Spatio-temporal Scales, IISc Bengaluru (Jan 2024)**: Poster Presentation – Best Poster Award

LEADERSHIP & ACTIVITIES

- Program Coordinator (Volunteer)**: Coordinated an experiential Himalayan trek for IIM Indore students, fostering leadership skills amidst challenging terrain and weather at 12,250 ft (Brahmatal, Uttarakhand)